

JPE Replication Report

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1 Paper Identification

ID	20250178
Author	Yang
Title	Appropriate Entrepreneurship? The Rise of China and the Developing World
Iteration	2

Some useful information up front:

- [JPE Data and Code Availability Policy](#)
- [Step by step guidance from Data Editor for Authors](#)
- [Template README](#)

2 Summary

In assessing compliance with our [Data and Code Availability Policy](#), our reproducibility team has identified the following issues, which we ask you to address:

1. Please look at our section 8 below and incorporate the specific requirement to reserve enough disk space for the procedure to run.
2. There was also a missing stata package, please add to the readme.

3 General

- ☒ Data Availability and Provenance Statements
 - ☒ Statement about Rights - Part 1: Right to use the data (e.g. *did the authors lawfully obtain the data*)
 - ☒ Statement about Rights - Part 2: Right to publish the data (e.g. *are human subjects involved and did permission been granted to publish their data?*)
 - ☒ License for Data (optional, but recommended)
 - ☒ Details on each Data Source
- ☒ Dataset list
- ☒ Computational requirements
 - ☒ Software Requirements
 - ☒ Controlled Randomness (as necessary)
 - ☒ Memory, Runtime, and Storage Requirements
- ☒ Description of programs/code
 - ☒ License for Code (Optional, but recommended)
- ☒ {REPLICATOR} to Replicators
 - ☒ Details (as necessary)
- ☒ List of tables and programs
- ☒ References (Optional, but recommended)

4 High-level Description of Research Project

This research project...

- ☒ uses observational data.
- ☒ uses *confidential* observational data.
- ☐ uses data generated via experiment or trial by the authors (in field or lab).
 - ☐ For lab experiments: The code to implement the experimental protocol (z-Tree, o-Tree etc) is included in the package.
 - ☐ A PDF copy of IRB approval is contained in the package.
 - ☐ A PDF document outlining the design of the experiment, including information on the selection and eligibility of participants is included.
 - ☐ A PDF copy of the instructions given to participants in both original language and an English translation is included.
- ☐ uses data generated via survey by the authors (in field or online)
 - ☐ The programs used to administer the survey (e.g. qualtrics survey file) are included.
 - ☐ A PDF copy of IRB approval is contained in the package.
 - ☐ A PDF of the original questionnaire(s) and information on the selection and eligibility of participants is included.
- ☐ is a simulation study: the data are generated by an algorithm.

☐ is a numerical illustration: no data are used but code is used to produce illustrative output (tables, figures)

5 Data description

For any data that cannot be provided as part of the replication package, please provide the following information as part of the README.

{REQUIRED} Please specify how long the data will be preserved in the restricted-access location.

{REQUIRED} Please provide affirmation of support for replication checks. - As per the [JPE policy](#), "In cases where data cannot be published in an openly accessible trusted data repository like the JPE dataverse, authors who have requested an exemption to publish them at the time of first submission must commit to preserving the data and code for a period of no less than five years following the publication of the paper. They should also provide reasonable assistance to requests for clarification and replication."

{REQUIRED} Please provide a codebook for the data. - The codebook should at a minimum describe the variables obtained from the data source, in a manner that will let others verify that they have obtained a substantially similar dataset if successfully obtaining access to the data.

5.1 Data Sources

file/name	public	private	DOI/URL	Access description
suitability_resource/WDI_full.xlsx	yes	no	https://databank.worldbank.org/source/world-development-indicators (CC-BY 4.0)	no
patent_resource/g_patent.tsv	yes	no	https://patentsview.org/download/data-download-tables (CC-BY 4.0)	yes
patent_resource/g_us_patent_citation.tsv	yes	no	https://patentsview.org/download/data-download-tables (CC-BY 4.0)	yes
patent_resource/g_cpc_current.tsv	yes	no	https://patentsview.org/download/data-download-tables (CC-BY 4.0)	yes
patent_resource/g_application.tsv	yes	no	https://patentsview.org/download/data-download-tables (CC-BY 4.0)	yes
patent_resource/g_assignee_disambiguated.tsv	yes	no	https://patentsview.org/download/data-download-tables (CC-BY 4.0)	yes
patent_resource/g_location_disambiguated.tsv	yes	no	https://patentsview.org/download/data-download-tables (CC-BY 4.0)	yes
patent_resource/g_inventor_disambiguated.tsv	yes	no	https://patentsview.org/download/data-download-tables (CC-BY 4.0)	yes
geocoding_resource/ne_10m_admin_0_countries/ (shapefile set)	yes	no	https://www.naturalearthdata.com (Admin 0–Countries; Populated Places. 1:10m Cultural Vectors)	no
geocoding_resource/ne_10m_populated_places. {shp,dbf,shx,prj,cpg}	yes	no	https://www.naturalearthdata.com (Admin 0–Countries; Populated Places. 1:10m Cultural Vectors)	no
geocoding_resource/simplemaps.csv	yes	no	https://simplemaps.com/data/world-cities (CC-BY 4.0)	no
geocoding_resource/opendatasoft.csv	yes	no	https://public.opendatasoft.com/explore/dataset/geonames-all-cities-with-a-population-1000/ (CC-BY 4.0)	no
other_data_resource/un_polity_china.dta (+ twin)	yes	no	https://doi.org/10.7910/DVN/LEJUQZ	no
other_data_resource/Trade_Volume_Export_Import.csv	yes	no	https://comtradeplus.un.org.	no
other_data_resource/table1_panelB_funding_sources.csv	yes	no	Not provided	no
pitchbook/pbdeals.dta	no	yes	Not provided	yes
pitchbook/pbcompanies.dta	no	yes	Not provided	yes
pitchbook/pbinvestors.dta	no	yes	Not provided	yes

file/name	public	private	DOI/URL	Access description
pitchbook/pbdealinvestor.dta	no	yes	Not provided	yes
pitchbook/pbmarketmap_v2.csv	no	yes	Not provided	yes
pitchbook/pbdealinvestor.dta	no	yes	Not provided	yes
pitchbook/later_years_extension/All_VC_Comp_Deal_Investor_2022-2024.csv	no	yes	Not provided	yes
pitchbook/later_years_extension/pbmarketmap_22_24.csv	no	yes	Not provided	yes
other_data_resource/LP.dta	no	yes	Not provided	yes

6 Requirements

- ✓ README is in TXT, MD, or PDF format
- ✓ README is accessible at the root of the package (not inside a folder)
- ✓ Title conforms to guidance (starts with “Data and Code for: Title of Paper” or “Code for: Title of Paper”, is properly capitalized)
- ✓ Authors (with affiliations) are listed in the same order as on the paper

6.1 Stated Requirements

- ☐ No requirements specified
- ✓ Software Requirements specified as follows:
 - Stata 19.5
 - Python >= 3.11
- ✓ Computational Requirements specified as follows:
 - At least 64gb of memory, parallel computing capabilities.
- ✓ Time Requirements specified as follows:
 - If parallel computing: ~8-10 hours. If not, could be up to 48 hours.
- ✓ Requirements are complete.

7 Analyzing Data and Code Files

7.1 Potential Personal Identifiable Information (PII)

⚠ We found the following instances of potentially personally identifying information. This may be completely legitimate but might be worth checking. As a reminder, privacy legislation in many countries (e.g. GDPR in EU) prohibits the dissemination of personal identifiable information without prior (and documented) consent of individuals. If indeed you want to publish such information with your replication package, you should probably have obtained IRB approval for this - please check!

Summary:

- Data files with PII indicators: 0
- Variables flagged in data: 0
- Code files with PII references: 113
- PII references in code: 5345

7.1.1 Summary of Flagged Files

File Type	File	Variables/References	PII Categories
Code	00_run_all.py	4	name
Code	01_generate_analysis_v2.do	97	loc, lat, country, name, lname, lon, location
Code	01_prepare_manifest.py	1	name
Code	02_generate_suitability.py	83	country, loc, name, zip, lat, lon, block
Code	02_train_bert_categories.py	16	son, lat, lname, name

File Type	File	Variables/References	PII Categories
Code	03_predict_companies.py	7	lname, name
Code	03a_generate_populated_places.do	9	loc, lat, city, lon, name
Code	03b_generate_controls.py	20	country, loc, name, lon, phone, lat
Code	03c_generate_regression_auxiliary.do	133	loc, country, lname, name, second, lat, location, lon
Code	04_generate_regression_corrected_120623.do	40	loc, lat, country, lname, name, block
Code	04_make_predicted_positive.py	14	name, lname
Code	05_make_company_level_predictions.py	13	country, name, lname
Code	05a_generate_postregression_inputs.do	22	loc, country, name, block
Code	05b_generate_suitability_gdp_component.do	9	loc, country
Code	05c_generate_regression_alltype_by_cat.do	63	loc, second, lat, country, name, lname, lon, block
Code	05d_generate_dealcount_alltype_cluster.do	30	loc, lname, name, country, lon
Code	05e_generate_pre_period_deals.do	7	loc, lname, name, country
Code	05f_generate_similarity_western.do	21	loc, name, lname, country
Code	05g_generate_regression_validation_countrypair.do	88	loc, country, name, lat, lon, lname
Code	05h_generate_deal_22_24.do	67	loc, lat, name, lname, country, lon

See [Appendix](#) for detailed listing of all flagged instances.

7.1.2 File Paths Report

Generated on 2026-09-03T14:19:40.454

Warning: Our search on file path types is imperfect and incurs both type 1 and type 2 errors. We aim to strike a reasonable balance between both. The below table is therefore only indicative. Detailed listings can be found in the appendix to this report.

In this table we analyse all files which contain source code (not data and documentation), and we report the grand total of each type of filepath we encountered.

Please check and replace any `windows` filepaths with unix compliant paths, insofar as this is possible in your setup. (Notice that `STATA` allows `/` to be a filepath separator on a windows platform, hence this is a requirement for *all* `STATA` applications.)

Files Analyzed	Windows Paths	Unix Paths	Mixed Paths	Drive Letters (<code>C:\</code> etc)
115	12	76	7	2

7.1.3 Duplicate Files Report

We did not find any duplicate files.

7.1.4 Zero Size Files Report

We did not find any zero sized files.

7.1.5 Large Files Report

We did not find any files larger than 100MB.

7.2 Code description

- Stage A prepares data.

- Each sub script in stage B generates one exhibit.

cloc github.com/AIDanial/cloc v 2.06 T=0.07 s (1665.2 files/s, 409063.8 lines/s)

Language	files	blank	comment	code
Stata	66	1223	9	7610
Python	33	716	149	3817
Jupyter Notebook	12	0	13590	1747
Markdown	3	76	0	384
Bourne Shell	4	41	10	318
Text	3	0	0	35
---	---	---	---	---
SUM:	121	2056	13758	13911

☒ The replication package contains a "main" or "master" file(s) which calls all other auxiliary programs.

NOTE: In-text numbers that reference numbers in tables do not need to be listed. Only in-text numbers that correspond to no table or figure need to be listed.

7.3 Computing Environment of the Replicator

- Onyxia [RStudio](#) instance with 25000m CPU allocation and 124 Gb of memory, 100Gb of persistent storage. Custom Docker image ([see here](#)) based on [dataeditors/stata19_5-mp](#)
- Running Debian 6.12.86-1 on [Intel\(R\) Xeon\(R\) Gold 6448Y](#) with 32 cores.
- Stata/MP 19.5
- Python 3.12 in a clean virtual environment

8 Replication

- Downloaded code and data from dropbox.
- Committed code to git repo.
- Used a custom [stata19_5-mp](#) Docker image to run on the instance.
- Installed python's [venv](#) package and activated a new virtual environment at the root of the filesystem in Onyxia service (as per [README](#) recommendations)
- Ran [code/run_all.sh](#)
- Stage A produces one error, on step [9](#):

```
>>> stata: code/generate/09_generate_simulated_deals.do
ERROR: code/generate/09_generate_simulated_deals.do hit a Stata error r(693); - see output/logs/generate/09_generate_simulated_deals.
```

- Logs show that this is due to lack of space on device.
- Deleting intermediate datasets from previous steps (once succesfully ran) provided enough space for this step to succeed.
- Some steps in Stage B did not succeed:
 - Two stage B steps failed:

```
>>> stata: code/figures/figure_A14.do
ERROR: code/figures/figure_A14.do hit a Stata error r(601); - see output/logs/figure_A14.log
>>> stata: code/figures/figure_A15.do
ERROR: code/figures/figure_A15.do hit a Stata error r(601); - see output/logs/figure_A15.log
```

- Due to missing package [gtools](#) which was supposed to be installed in the package installation script. The package could be installed with [ssc install gtools](#).
- After fixing issues mentionned above, all exhibits were generated.

9 Findings

9.1 Data Preparation Code

9.2 Tables and Figures

Data Preparation Program	Dataset created	Reproduced?	
Files in code/BERT_generation	9 files in BERT_prediction_resource/ (+ CSV twins)	Yes	
Figure/Table #	Program	Line Number	Reproduced?
Table 1	code/tables/table_1.py	-	Yes
Table 2	code/tables/table_2.do	-	Yes
Table 3	code/tables/table_3.do	-	Yes
Table 4	code/tables/table_4.do	-	Yes
Table 5	code/tables/table_5.do	-	Yes
Table 6	code/tables/table_6.do	-	Yes
Table 7	code/tables/table_7.do	-	Yes
Table 8	code/tables/table_8.do	-	Yes
Table A1	code/tables/table_A1.py	-	Yes
Table A3	No script (textual data)	-	-
Table A2	code/tables/table_A2.do	-	Yes
Table A4	code/tables/table_A4.do	-	Yes
Table A5	code/tables/table_A5.do	-	Yes
Table A6	code/tables/table_A6.do	-	Yes
Table A7	code/tables/table_A7.do	-	Yes
Table A8	code/tables/table_A8.do	-	Yes
Table A9	code/tables/table_A9.do	-	Yes
Table A10	code/tables/table_A10.do	-	Yes
Table A11	code/tables/table_A11.do	-	Yes
Table A12	code/tables/table_A12.do	-	Yes
Table A13	code/tables/table_A13.do	-	Yes
Table A14	code/tables/table_A14.do	-	Yes
Table A15	code/tables/table_A15.do	-	Yes
Table A16	code/tables/table_A16.do	-	Yes
Table A17	code/tables/table_A17.do	-	Yes
Table A18	code/tables/table_A18.do	-	Yes
Table A19	No script (textual data)	-	-
Table A20	code/tables/table_A20.do	-	Yes
Table A21	code/tables/table_A21.do	-	Yes
Table A22	code/tables/table_A22.do	-	Yes
Table A23	code/tables/table_A23.do	-	Yes
Table A24	code/tables/table_A24.do	-	Yes

Data Preparation Program	Dataset created	Reproduced?	
Table A25	code/tables/table_A25.do	-	Yes
Table A26	code/tables/table_A26.do	-	Yes
Table A27	code/tables/table_A27.do	-	Yes
Table A28	code/tables/table_A28.do	-	Yes
Table A29	code/tables/table_A29.do	-	Yes
Table A30	code/tables/table_A30.do	-	Yes
Figure 1	code/figures/figure_1.py	-	Yes
Figure 2	code/figures/figure_2.do	-	Yes
Figure 3	code/figures/figure_3.do	-	Yes
Figure 4	code/figures/figure_4.do	-	Yes
Figure A1	code/figures/figure_A1.do	-	Yes
Figure A2	code/figures/figure_A2.do	-	Yes
Figure A3	code/figures/figure_A3.do	-	Yes
Figure A4	code/figures/figure_A4.do	-	Yes
Figure A5	code/figures/figure_A5.do	-	Yes
Figure A6	code/figures/figure_A6.do	-	Yes
Figure A7	code/figures/figure_A7.do	-	Yes
Figure A8	code/figures/figure_A8.do	-	Yes
Figure A9	code/figures/figure_A9.do	-	Yes
Figure A10	code/figures/figure_A10.do	-	Yes
Figure A11	code/figures/figure_A11.do	-	Yes
Figure A12	code/figures/figure_A12.do	-	Yes
Figure A13	code/figures/figure_A13.do	-	Yes
Figure A14	code/figures/figure_A14.do	-	Yes
Figure A15	code/figures/figure_A15.do	-	Yes
Figure A16	code/figures/figure_A16.do	-	Yes
Figure A17	code/figures/figure_A17.do	-	Yes
In-text numbers	Program	Line Number	

9.3 In-Text Numbers

- ☐ In-text numbers not verified.
- ☒ There are no in-text numbers, or all in-text numbers stem from tables and figures.
- ☐ There are in-text numbers, but they are not identified in the code

9.4 Other Issues

10 Classification

- ☐ full reproduction
- ☒ full reproduction with minor issues

- ☐ partial reproduction (see above)
- ☐ not able to reproduce most or all of the results (reasons see above)

10.1 Reason for incomplete reproducibility

- ☒ None.
- ☐ **Discrepancy in output** (either figures or numbers in tables or text differ)
- ☐ **Bugs in code** that were fixable by the replicator (but should be fixed in the final deposit)
- ☐ **Code missing**, in particular if it prevented the replicator from completing the reproducibility check
- ☐ **Data preparation code missing** should be checked if the code missing seems to be data preparation code
- ☐ **Code not functional** is more severe than a simple bug: it prevented the replicator from completing the reproducibility check
- ☐ **Software not available to replicator** may happen for a variety of reasons, but in particular (a) when the software is commercial, and the replicator does not have access to a licensed copy, or (b) the software is open-source, but a specific version required to conduct the reproducibility check is not available.
- ☐ **Insufficient time available to replicator** is applicable when (a) running the code would take weeks or more (b) running the code might take less time if sufficient compute resources were to be brought to bear, but no such resources can be accessed in a timely fashion (c) the replication package is very complex, and following all (manual and scripted) steps would take too long.
- ☐ **Data missing** is marked when data *should* be available, but was erroneously not provided, or is not accessible via the procedures described in the replication package
- ☐ **Data not available** is marked when data requires additional access steps, for instance purchase or application procedure.
- ☐ **Missing README** is marked if there is no README to guide the replicator, or the README is not in compliance with JPE requirements

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Code	02_generate_suitability.py	83	country, loc, name, zip, lat, lon, block
Code	02_train_bert_categories.py	16	son, lat, lname, name
Code	03_predict_companies.py	7	lname, name
Code	03a_generate_populated_places.do	9	loc, lat, city, lon, name
Code	03b_generate_controls.py	20	country, loc, name, lon, phone, lat
Code	03c_generate_regression_auxiliary.do	133	loc, country, lname, name, second, lat, location, lon
Code	04_generate_regression_corrected_120623.do	40	loc, lat, country, lname, name, block
Code	04_make_predicted_positive.py	14	name, lname
Code	05_make_company_level_predictions.py	13	country, name, lname

File Type	File	Variables/References	PII Categories
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Code	05b_generate_suitability_gdp_component.do	9	loc, country
Code	05c_generate_regression_alltype_by_cat.do	63	loc, second, lat, country, name, lname, lon, block
Code	05d_generate_dealcount_alltype_cluster.do	30	loc, lname, name, country, lon
Code	05e_generate_pre_period_deals.do	7	loc, lname, name, country
Code	05f_generate_similarity_western.do	21	loc, name, lname, country
Code	05g_generate_regression_validation_countrypair.do	88	loc, country, name, lat, lon, lname
Code	05h_generate_deal_22_24.do	67	loc, lat, name, lname, country, lon

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